Toward a four-dimensional Yang–Mills construction

Notebook 06: direct flat Maxwell bounds and a compact lift

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Abstract

This note proves bounds for the flat quadratic effective action that are uniform in the number of averaging levels. Curvature averaging is a contraction in the correctly scaled volume norms. A finite-block Poincaré argument therefore supplies an axial gap. An explicit compact right inverse to straight-string averaging supplies the complementary energy upper bound. A gauge adjustment shows that the rooted and straight constraints have the same energy infimum. Together with the actual averaged-path gauge bridge of Notebook 05, these results give both bounds for the flat reduced Gaussian precision, subject to the stated identification with the constrained-minimizer Hessian. Complex background responses and covariance locality remain separate problems.

Status. The proofs have passed separate internal checks of the averaging, finite-block gap, compact lift, periodic counting, and variational transfer. No external certification or novelty claim is made. The statements below are about an explicitly defined flat quadratic model. Section 6 gives the precise additional identification needed for the deposited Yang–Mills Gaussian; no nonlinear renormalization theorem or continuum limit is inferred from these bounds.

1 Normalized averaging commutes with curvature

Let fine and coarse spacings be h and $H = Mh$, with M odd. Use centered M-blocks $B_M(y)$, one positive bond orientation, and one plaquette orientation $\mu < \nu$. Potentials are physical coordinates: the group link would be $\exp(ihA_\mu(x))$. Define

$$(d_h A)_{\mu\nu}(x) = h^{-1}[A_\nu(x + he_\mu) - A_\nu(x) - A_\mu(x + he_\nu) + A_\mu(x)],$$

$$(Q_M^{\mathrm{st}} A)_\mu(y) = M^{-d-1} \sum_{x \in B_M(y)} \sum_{r=0}^{M-1} A_\mu(x + rhe_\mu),$$

$$(Q_M^{(2)} F)_{\mu\nu}(y) = M^{-d-2} \sum_{x \in B_M(y)} \sum_{r,s=0}^{M-1} F_{\mu\nu}(x + rhe_\mu + she_\nu).$$

The volume norm is $\|F\|_h^2 = h^d \sum_x \sum_{\mu<\nu} |F_{\mu\nu}(x)|^2$, and analogously for one-forms.

Proposition 1.1. *On the infinite lattice or a periodic lattice tiled by complete blocks,*

$$d_H Q_M^{\mathrm{st}} = Q_M^{(2)} d_h, \qquad \left\| Q_M^{(2)} F \right\|_H \le \|F\|_h . \tag{1}$$

The one-form average is also a contraction in the volume norms.

Proof. Expand a coarse plaquette curl and translate the block base points. The four strings are the boundary of an M by M fine plaquette square. Discrete Stokes contributes $h\sum d_h A$, and the coarse curl contributes $H^{-1} = (Mh)^{-1}$. Together with M^{-d-1}, this gives exactly M^{-d-2}.

Each row of $Q_M^{(2)}$ is a positive average of M^{d+2} terms. For a fixed fine plaquette and each pair (r, s), the base point lies in exactly one block. Its total column weight is therefore $M^2 M^{-d-2} = M^{-d}$. Jensen's inequality, followed by $H^d M^{-d} = h^d$, proves the contraction. Periodic preimages are counted with multiplicity, so small tori cause no exception. Use M string indices in place of M^2 surface indices for the one-form contraction. □

The physical derivative h^{-1} is essential in this computation. Unscaled plaquette sums would change the power of M. Straight-string averages satisfy the exact semigroup identity

$$Q_N^{\mathrm{st}} Q_M^{\mathrm{st}} = Q_{NM}^{\mathrm{st}}.$$

Indeed a centered NM-block decomposes uniquely into centered M-blocks with intermediate base coordinates, and a string index decomposes uniquely as $r_M + M r_N$. The two bijections are independent. The same proof gives the two-form semigroup identity.

1.1 The rooted contour average

Let $T_h A(y, x)$ be the physical rooted path integral, including the factor h, and let a average it over the block. The linearization of the rooted group average is

$$Q_M^{\flat} = Q_M^{\mathrm{st}} - D_H a T_h. \tag{2}$$

The initial and final rooted contours give the second term; the reference coarse string cancels with the final reference transporter. This is the spacing-restored identity of Notebook 05. Since $d_H D_H = 0$, the curvature identity (1) also holds with $Q_M^{\flat}$. Iterating gives

$$\|d_1 Q_k A\|_1 \le \|d_\eta A\|_\eta, \qquad \eta = L^{-k}, \tag{3}$$

for the composed rooted derivative. No equality of its one-form operator with $Q_{L^k}^{\mathrm{st}}$ is used.

The corrected operator in Raw I, lines 24355–24375, has a different flat derivative: all four correction brackets vanish at the identity, leaving Q^{st} exactly. Its straight semigroup is valid. The rooted operator of CMP109 differs by the gradient in (2); the two should not be identified merely by notation.

2 A compact right inverse with bounded energy

Set coarse spacing to one and fine spacing to $\eta = 1/M$. We now construct a right inverse R_M to Q_M^{st} with constants independent of M. This section uses infinite or periodic complete block lattices. Restriction to an arbitrary boundary is discussed below.

For $M = 2h_0 + 1 \ge 3$, put

$$m_1 = \frac{M^2 - 1}{4M^2}, \qquad m_2 = \frac{M^2 - 1}{12M^2}, \qquad v = m_2 - m_1^2 = \frac{(M^2 - 1)(M^2 + 3)}{48M^4}.$$

Define a longitudinal sequence, zero outside $0 \le s < M$, by

$$u_s = (s - h_0)/M, \qquad \psi_M(s) = \alpha + \beta |u_s|, \qquad \alpha = m_2/v, \quad \beta = -m_1/v. \tag{4}$$

For the normalized mean $E_M = M^{-1} \sum_{s=0}^{M-1}$,

$$E_M \psi_M = 1, \qquad E_M(|u|\psi_M) = 0, \qquad E_M \psi_M^2 = \alpha \le 4. \tag{5}$$

These follow by substituting $E_M|u| = m_1$, $E_M u^2 = m_2$. In particular the signed sequence is even about h_0.

For transverse coordinates define the hat sequence

$$\phi_M(r) = \frac{M}{h_0+1}\left(1 - \frac{|r|}{h_0+1}\right) \quad (|r| \le h_0), \qquad \phi_M(r) = 0 \quad \text{otherwise.} \tag{6}$$

Direct summation gives

$$M^{-1}\sum_r \phi_M(r) = 1, \quad 0 \le \phi_M \le 2, \quad M^{-1}\sum_r \phi_M(r)^2 \le 2, \tag{7}$$

$$M^{-1}\sum_r |M[\phi_M(r+1) - \phi_M(r)]|^2 = \frac{16M^3}{(M+1)^3} \le 16. \tag{8}$$

There are $M+1$ nonzero differences, all with absolute value $M/(h_0+1)^2$ before multiplying by M.

For a coarse bond based at $j \in \mathbb{Z}^d$ in direction μ, define the fine bond basis, writing $z = Mx \in \mathbb{Z}^d$, by

$$\Psi_{\mu,j}(z) = \psi_M(z_\mu - Mj_\mu)\prod_{\nu\neq\mu}\phi_M(z_\nu - Mj_\nu),$$

and set

$$(R_M B)_\mu(z/M) = \sum_j B_\mu(j)\Psi_{\mu,j}(z). \tag{9}$$

The sums are local. Each basis reads only its own coarse bond and is supported in the union of the two endpoint blocks. Periodize the basis for a torus. For $M = 1$, use the identity lift.

Theorem 2.1 (Compact bounded-energy lift)**.** *For M odd and $d \ge 2$,*

$$Q_M^{\mathrm{st}} R_M = I, \qquad \|R_M B\|_\eta^2 \le 2^{d+1}\,\|B\|_1^2, \tag{10}$$

$$\|d_\eta R_M B\|_\eta^2 \le C_d\,\|B\|_1^2, \qquad C_d = 2^{d+6}(d-1). \tag{11}$$

The bounds hold in infinite volume and on periodic lattices, including small tori with periodic multiplicities.

Proof. The averaging separates into coordinates. Transversely, the supports of $\phi_M(r - Mj)$ are disjoint blocks and their block averages are Kronecker deltas by (7). Longitudinally, the straight-string kernel in a row based at Mj is the triangle

$$M^{-1}\left(1 - \frac{|s - Mj - h_0|}{M}\right)_+,$$

on its integer support. On the support of the basis with index j, the target row thus averages $(1-|u|)\psi_M$, which is one by (5). The adjacent rows average the positive and negative halves of $|u|\psi_M$. Both are zero by evenness and the second moment condition. Every other row has disjoint support. This proves $QR = I$; periodization preserves the identity.

For each orientation the tensor supports tile the fine sites without overlap. The squared norm of a basis is bounded by $4\,2^{d-1} = 2^{d+1}$, using the volume-weighted tensor product of (5) and (7). This proves the first norm bound.

Only transverse derivatives $D_{\eta,\nu}A_\mu$, $\nu \neq \mu$, enter the curl. Thus no derivative of the signed longitudinal sequence is needed. For one basis, its transverse derivative has squared norm at most $4\,16\,2^{d-2} = 2^{d+4}$. At a given fine bond, at most two differentiated translates overlap. Cauchy–Schwarz therefore gives

$$\|D_{\eta,\nu}(R_M B)_\mu\|_\eta^2 \le 2^{d+5}\,\|B_\mu\|_1^2.$$

Finally sum $|D_\mu A_\nu - D_\nu A_\mu|^2 \le 2|D_\mu A_\nu|^2 + 2|D_\nu A_\mu|^2$ over $\mu < \nu$. Each component occurs $d-1$ times, proving (11). The same overlap counting on the periodic cover proves the bound for small tori. The identity lift at $M = 1$ obeys the displayed conservative constants directly. □

For example, at $M = 3$, $\psi = (0, 3, 0)$ and $\phi = (3/4, 3/2, 3/4)$. At larger M, the longitudinal basis has negative portions; that cancellation is what makes it an exact right inverse. Positivity of the lift is neither asserted nor needed.

For supported coarse fields on the full lattice this lift remains supported in the corresponding double blocks. A coarse bond incident to a bounded region can have one endpoint block outside it. The unmodified basis then needs a collar. The following separate construction removes that need for the specified incident-bond boundary convention.

2.1 An exact incident-bond boundary lift

Let S be any collection of coarse sites and let the fine region be the union of their complete centered M-blocks. Retain bonds with at least one endpoint in the corresponding region; set every other coarse or fine coordinate to zero. The constraint is the full straight-string average of this zero-extended field, with no truncation or renormalization of boundary rows. The energy is the full curl norm of the zero extension.

For $M = 2h_0 + 1 \ge 3$, put

$$a_M = \frac{2M^2}{(h_0+1)(h_0+2)} \le 8.$$

Replace the longitudinal profile on an outgoing coarse bond (initial site in S, final site outside) by

$$\psi_M^{\mathrm{out}}(s) = a_M[1 - 3(h_0 - s)/M] \quad (0 \le s \le h_0),$$

zero elsewhere. For an incoming positive bond use its reflection

$$\psi_M^{\mathrm{in}}(s) = a_M[1 - 3(s - h_0)/M] \quad (h_0 \le s < M),$$

zero elsewhere. Keep the original profile on bonds with both endpoints inside. The interface bond $s = h_0$ is retained in either boundary case, and is included only once in the selected profile.

For $p(r) = a_M(1 - 3r/M)$, $0 \le r \le h_0$, direct summation gives

$$\frac{1}{M}\sum_r p(r) = 1, \qquad \frac{1}{M}\sum_r \frac{r}{M} p(r) = 0, \qquad \frac{1}{M}\sum_r p(r)^2 = a_M \le 8.$$

Indeed $\sum r = h_0(h_0+1)/2$ and $\sum r^2 = h_0(h_0+1)M/6$ prove the first two equalities; multiplying $p = a_M(1 - 3r/M)$ by p proves the third. The target triangular averaging row therefore remains one. One neighboring row vanishes by the first-moment identity, and the other has disjoint support. Every further row vanishes. Thus every selected profile still maps to its exact single coarse bond.

An outgoing profile uses only fine bonds inside the initial block and its crossing interface bond. An incoming profile uses the crossing bond and bonds inside the final block. With the unchanged transverse hats, every used fine bond has an endpoint in the allowed region, even when S has holes. Tensoring and summing as in (9) defines R_M^S with

$$Q_M^{\mathrm{st}} R_M^S B = B, \qquad \left\| R_M^S B \right\|_\eta^2 \le 2^{d+2} \|B\|_1^2, \tag{12}$$

$$\left\| d_\eta R_M^S B \right\|_\eta^2 \le 2^{d+7}(d-1) \|B\|_1^2 .$$

The constants are independent of the region and refinement. The proof of Theorem 2.1 applies with longitudinal squared norm eight in place of four. In particular, differentiate each fixed basis before summing: even when adjacent source bonds use different profile types, only two transverse derivative supports overlap. No derivative of a position-dependent profile choice is introduced. The same argument works on periodic covers, and $M = 1$ uses the identity. In four dimensions the boundary energy constant is 6144.

The identity holds on all coarse rows, including the excluded ones. Indeed an excluded row's strings lie in its two exterior endpoint blocks and read no allowed fine bond. Thus full-row and incident-row constraints agree for this convention. The one-step deposited flat convention matches: Raw II, lines 9032–9039, uses incident bonds; the positivity setup beginning at line 9743 uses complete blocks; lines 9779–9791 use Q_1, whose flat derivative is Q^{st}.

This proves the boundary upper bound for that exact linear problem. It does not identify an arbitrary multilevel retained-history minimizer, establish a boundary covariance estimate, or cover a convention freezing crossing bonds. Nonlinear transfer still needs the corresponding constraint and gauge match of Section 6.

2.2 An independent periodic Fourier upper bound

There is a separate check of scale uniformity. For a coarse Fourier mode $k \in [-\pi, \pi]^d$, embed the same physical frequency in the fine torus. Let

$$r_M(k) = \frac{\sin(k/2)}{M \sin(k/(2M))}, \qquad r_M(0) = 1.$$

For centered blocks the averaging multiplier is

$$q_\mu(k) = e^{i(M-1)k_\mu/(2M)} r_M(k_\mu) \prod_{\nu=1}^{d} r_M(k_\nu).$$

Since $|r_M(k)| \geq 2/\pi$ on this zone, divide the component coefficient by $q_\mu(k)$ to obtain a right inverse. Fine-volume Parseval equals coarse Parseval for the embedded frequencies. The exterior derivative multiplier has squared norm at most $\sum_\nu |M(e^{ik_\nu/M} - 1)|^2 \leq d\pi^2$. Hence

$$\inf_{Q_M^{\mathrm{st}} A = B} \|d_\eta A\|_\eta^2 \leq d\pi^2 (\pi/2)^{2d+2} \|B\|_1^2 .$$

For real B, taking the real part of a complex lift preserves the constraint and cannot increase energy, covering Nyquist conventions. Unlike (9), this Fourier lift is not compactly supported.

3 The flat variational form and rooted constraints

Define, first as a standalone quadratic model,

$$q_M(B) = \inf_{Q_M^{\mathrm{st}} A = B} \|d_\eta A\|_\eta^2 , \qquad \eta = M^{-1}. \tag{13}$$

The constraint is nonempty by Theorem 2.1. On a finite torus this is a nonnegative quadratic form: it is least squares on a nonempty affine space. Equations (1) and (11) give directly

$$\|d_1 B\|_1^2 \leq q_M(B) \leq C_d \|B\|_1^2 . \tag{14}$$

It is invariant under adding a coarse gradient. Indeed $Q_M^{\mathrm{st}} D_\eta \lambda = D_1 \overline{\lambda}$, where $\overline{\lambda}$ is the scalar block average; this average is onto, and gradients have zero curl energy.

The composed rooted constraint has the same infimum. In one step,

$$Q_M^{\flat} D_\eta \lambda = D_1(\lambda \text{ at centers}), \qquad Q_M^{\mathrm{st}} D_\eta \lambda = D_1 \overline{\lambda}.$$

Equation (2) and induction therefore give

$$Q_k^{\mathrm{rooted}} A = Q_{L^k}^{\mathrm{st}} A + D_1 \Xi_k(A)$$

for a linear scalar map Ξ_k. If the straight constraint equals B, extend the coarse-center values $-\Xi_k(A)$ to a fine site field and add its gradient. Nested center restriction corrects the rooted constraint to B, without changing curl energy. Conversely, the scalar block average has a right inverse (blockwise constants), so a gradient also corrects a rooted-feasible field to a straight-feasible one. Both infima are equal. This uses no bound on Ξ_k and proves no bound on the gauge-adjusted potential or its cutoff coordinates.

4 A finite-block axial Poincaré inequality

Fix the one-step blocking factor $L \geq 3$, independent of $M = L^k$. On the unit lattice let Ω consist of the 3^d adjacent L-blocks indexed by $\{-1, 0, 1\}^d$. Include every bond and plaquette internal to this rectangular grid. Let $T_{0,\Omega}$ contain the root-to-site integrals for a chosen coordinate-ordered tree in every block, and let $Q_{\mathrm{st},\Omega}$ contain the straight-string rows between all adjacent block centers inside Ω. Each such string stays in its two endpoint blocks. Form the completely specified finite matrix

$$M_L = d_\Omega^* d_\Omega + T_{0,\Omega}^* T_{0,\Omega} + Q_{\mathrm{st},\Omega}^* Q_{\mathrm{st},\Omega}, \qquad \sigma(d, L) = \lambda_{\min}(M_L).$$

Lemma 4.1. *One has $\sigma(d, L) > 0$. Every global field with $T_0 B = 0$ and $Q_1 B = Q_L^{\mathrm{st}} B = 0$ satisfies*

$$\|d_1 B\|_1^2 \geq \gamma_{\mathrm{loc}} \|B\|_1^2, \qquad \gamma_{\mathrm{loc}} = \sigma(d, L)/3^d > 0. \tag{15}$$

The constant is independent of volume, averaging depth, and color dimension.

Proof. If the matrix quadratic form vanishes, all local plaquette curls vanish. Path independence on a rectangular grid gives $B = D\varphi$; interchanging adjacent coordinate steps is justified by each zero plaquette curl. The tree equations force φ to be constant within every block. A straight-string row equals the difference of the two block constants divided by L. The block graph is connected, so all constants agree, and $B = 0$. Thus M_L has trivial kernel, which proves strict positivity of its smallest eigenvalue.

Restrict a globally constrained field to each cube Ω_y translated by an aligned block center $y \in L\mathbb{Z}^d$. Its local tree and averaging rows vanish, giving

$$\sum_{b \subset \Omega_y} |B_b|^2 \leq \sigma(d, L)^{-1} \sum_{p \subset \Omega_y} |(d_1 B)_p|^2.$$

Every global bond occurs in at least one cube, and each plaquette in at most 3^d cubes. Summing proves (15). On a torus pull the field back to the ordinary cubes in the covering lattice. Even when a cube wraps, the same proof and counting with multiplicity apply. In infinite volume, use square-summable fields and monotone convergence. □

No numerical computation of σ is needed for strict positivity. The value may be small as a function of fixed d, L; its independence of the ultraviolet scale is the relevant conclusion. Together with (14), the lemma proves the full-field axial bound $q_M(B) \geq \gamma_{\mathrm{loc}} \|B\|^2$ directly.

5 Both bounds for the flat exponential-gauge precision

Use the averaged path map T, rooted constraint $Q^\flat$, and pivot insertion $K_\flat$ from Notebook 05. Set

$$d_0 = 2\sqrt{d}, \quad t_0 = \sqrt{d(L-1)(L^d-1)}, \quad r = 1 + 2d(L-1)/L,$$

$$K_*^2 = 1 + L^{2d}(2d+2)r^2, \qquad \gamma_* = \left((1+d_0t_0)^2/\gamma_{\mathrm{loc}} + d_0^2\right)^{-1}.$$

The constraint-compatible gauge bridge gives, on $\ker Q^\flat$,

$$q_M(B) + \|TB\|^2 \geq \gamma_* \|B\|^2 .$$

In fact the lower estimate needs only $q_M(B) \geq \|d_1 B\|^2$: the maps $B \mapsto B_T \mapsto B_0$ of Notebook 05 change fields by gradients, so their curls agree. Apply Lemma 4.1 to B_0, then $\|B_T\| \leq (1 + d_0 t_0)\|B_0\|$, and finally weighted Cauchy–Schwarz to $B = B_T + D(TB)$. Thus this particular lower bound does not need a separate gauge-invariance hypothesis on the effective quadratic form. The direct upper bound (14) applies on all bond fields, so also

$$q_M(B) + \|TB\|^2 \leq (C_d + t_0^2)\|B\|^2 .$$

With $B = K_\flat z$, $\|K_\flat\| \leq K_*$ and z a coordinate restriction of B, the reduced real precision satisfies

$$\gamma_* I \leq A_{M,0} \leq B_* I, \qquad B_* = (C_d + t_0^2)K_*^2. \tag{16}$$

All constants depend only on fixed block geometry and dimension in the displayed normalization. In particular,

$$B_*^{-1} I \leq C_{M,0} = A_{M,0}^{-1} \leq \gamma_*^{-1} I.$$

If only the Wilson quadratic action is multiplied by $c > 0$ while the gauge penalty stays $\|TB\|^2$, the appropriate constants are

$$\gamma_*(c) = \left((1+d_0t_0)^2/(c\gamma_{\mathrm{loc}}) + d_0^2\right)^{-1}, \qquad B_*(c) = (cC_d + t_0^2)K_*^2.$$

Rescaling both terms together instead rescales the whole precision.

For later use, if the actual complex symmetric precision is $A = A_{M,0} + V$ with $\|V\| \leq \gamma_*/2$, then

$$\operatorname{Herm} A \geq (\gamma_*/2)I, \qquad \|A\| \leq B_* + \gamma_*/2,$$

$$\operatorname{Herm} A^{-1} \geq \frac{\gamma_*/2}{(B_* + \gamma_*/2)^2} I, \qquad \left\|A^{-1}\right\| \leq 2/\gamma_*.$$

This last implication is elementary; the actual uniform perturbation estimate is not proved here. The spectral bounds alone do not give exponential covariance decay. Notebook 07 supplies that additional flat spatial estimate by an exact Fourier quotient and complex-pole cancellation. Localized background dependence remains a separate input.

6 Identification with the Yang–Mills flat Hessian

For the Wilson action in physical normalization,

$$S_\eta(U) = \eta^{d-4} \sum_p [1 - \Re \operatorname{tr} U(\partial p)],$$

and $U_b(t) = \exp(i\eta t A_b)$, stationarity at the identity and the first plaquette phase $i\eta^2 t(d_\eta A)_p$ give

$$D^2 S_\eta(1)[A, A] = \eta^d \sum_p |(d_\eta A)_p|^2.$$

The trace fixes the Lie-algebra norm. A different fixed normalization of the action multiplies its quadratic energy bounds by a positive factor; the preceding paragraph explains the gauge-penalty adjustment. Colors decouple at quadratic order.

Let $U_k(V)$ be the local full-block constrained minimizer with $U_k(1) = 1$ in its chosen gauge, and assume the branch is C^2 on a neighborhood of that datum. Write

$$U_k(e^{itB}) = \exp(i\eta[tL_k B + O(t^2)]).$$

The constraint gives $Q_k L_k B = B$. Since the fine action has zero gradient, its effective Hessian is

$$q_{\mathrm{eff},k}(B) = \|d_\eta L_k B\|_\eta^2 \geq \|d_1 B\|_1^2 .$$

The lower bound uses the derivative, constraint, and action expansion; the C^2 hypothesis makes it a statement about the classical effective Hessian. Mere differentiability would instead give the expansion $tL_k B + o(t)$ and the corresponding quadratic Peano coefficient. To identify the Hessian with the variational form q_M and obtain the upper bound, also use minimization against every tangent-feasible curve. At each fixed finite regulator, the linear average is onto by the lift and gradient adjustment above. For the actual analytic constraint, the finite-dimensional implicit function theorem therefore produces a feasible curve for every tangent-feasible field, provided the constraint map is defined on an open neighborhood of the identity. Comparing its action with the local minimizer and dividing by $t^2/2$ gives

$$q_{\mathrm{eff},k}(B) = \inf_{Q_k A = B} \|d_\eta A\|_\eta^2 = q_{L^k}(B).$$

The reverse inequality uses the feasible derivative $L_k B$ itself. If the chosen chart imposes an additional gauge, one must supply its constraint-preserving gauge equivalence before using every such competitor. The equality is not obtained from stationarity alone. No uniform nonlinear implicit-function radius is claimed by this finite-regulator argument.

CMP109, (0.21)–(0.22) and (2.6)–(2.11), identifies the effective action and the quadratic Gaussian in these terms. The deposited Raw II small-field step around lines 73870–74020 records the same minimizer variables and elimination. The first derivative of the group average must be checked from its contour definition, as done above; gauge covariance alone would not determine it. The corrected and rooted conventions have equal flat energy infima only through the explicit gradient adjustment proved here.

Notebook 15 now supplies the analytic ordinary-constraint local minimum and its comparison modulo gauge on the homogeneous small global potential class. Notebook 17 checks the preceding tangent comparison and deposited coordinate match explicitly. Thus (16) applies to that homogeneous flat source Gaussian. This does not apply automatically to domains with holes or mixed large-field histories, or identify a freely prescribed current with a physical action derivative. It does not transport the sharp cutoff under an untracked coordinate change.

For the quadratic cutoff coefficient evaluated at $(U, J) = (1, 0)$, this flat reference is the relevant starting point. What remains is a uniform weighted bound on its actual complex perturbations and source responses. Notebook 07 proves the homogeneous flat spatial decay, exact compatible-periodic volume identification, and a free refinement limit with $O(M^{-2})$ weighted error. These results remove the flat spectral and spatial estimates from the outstanding inputs; they do not complete the background neighborhood or a four-dimensional Yang–Mills theory.

References

[1] Pinned Yang–Mills-LLM-experiment account, commit `e392ef45013a581664bab76b908671 70b30fe10e`.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. 109 (1987), 249–301. Copy of the original article.